THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 06

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** it comes from the Latin word celeritas meaning speed	symbols come from history and agreement
2	**(c)** provides a tested explanation of why patterns occur	theory is evidence-based, not a hunch
3	**(d)** homogeneous	uniform composition
4	**(d)** a separating funnel	immiscible liquids form two layers
5	**(a)** loses one electron to form Na^+	completes octet by losing its single valence electron
6	**(d)** 98 u	2 + 32 + 64
7	**(d)** protons in its nucleus	defines the element
8	**(d)** zero	returns to start
9	**(b)** 25 m	s = ut + ½at²
10	**(a)** 6 N	add if same direction; subtract if opposite
11	**(d)** opposite to its motion	opposes relative motion
12	**(b)** joule	N m
13	**(a)** 85 m	distance = vt/2
14	**(d)** mitochondrion	ATP production
15	**(a)** Rudolf Virchow	1855
16	**(a)** sclerenchyma	hard and strong
17	**(d)** classification	taxonomy
18	**(c)** father (X or Y)	mother gives X always
19	**(c)** seed	contains the embryo
20	**(a)** 1 kW m^{-2}	less than the solar constant

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Weather depends on many changing factors and tiny differences grow over time, giving very different outcomes (2).	2
22	(a) carbon monoxide (b) dinitrogen pentoxide (c) hydrogen sulfide (d) sulfur hexafluoride (½ each).	½+½+½+½
23	v = u + at (½); s = ut + ½at² (1); v² = u² + 2as (½).	2
24	Forces which are equal and opposite on the same object so the net force is zero (1); a book at rest on a table (1).	1+1
25	More space for haemoglobin to carry oxygen (1); they cannot divide or repair, so they live only 120 days (1).	1+1
26	Skeletal, smooth and cardiac (2).	2
27	Sperm (male) and egg or ovum (female) (2).	2
28	Save energy, use public transport, plant trees, use renewable energy (1+1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Heterogeneous but appear homogeneous; particles do not settle; scatter light (Tyndall effect); pass through filter paper (any three) (2); e.g. milk, fog, smoke (1).	2+1
30	Weighted average $= 35 \times 0.75 + 37 \times 0.25$ (1) $= 26.25 + 9.25 = 35.5$ u (2).	1+2
31	Heat transfer from hot to cold bodies (1.5); radiation, e.g. the Sun's energy (1.5).	1½+1½
32	Microphone: sound makes a membrane vibrate, converted to an electrical signal (1.5); speaker: electrical signal makes a cone vibrate and produce sound (1.5).	1½+1½
33	Squamous (alveoli), cuboidal (kidney tubules), columnar (intestine), ciliated (trachea) (3).	3

34	Kingdom → Phylum → Class → Order → Family → Genus → Species (2); each level has more shared features, e.g. for humans, Animalia, Chordata, Mammalia, Primates (1).	2+1
35	(i) Transfer vs sharing of electrons (1); (ii) high vs low (1); (iii) conduct in molten/aqueous state vs generally non-conductors (1).	1+1+1
	OR: Their valence shell is complete (8 electrons or duplet for helium) so they are stable and do not combine (2); helium, neon, argon (1).	2+1
36	Plants fix CO_2 by photosynthesis (1); carbon passes through food chains and returns by respiration and decay (1); oceans store carbon (1).	1+1+1
	OR: Save energy, plant trees, use renewable energy, conserve water, sustainable farming (1+1+1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Na loses one electron, Cl gains it; the ions are held by electrostatic attraction (2). (ii) Na^+ (2,8), Cl^- (2,8,8) (1). (iii) MgO: Mg (2,8,2) loses 2 e^- to O (2,6) forming Mg^{2+} and O^{2-} (2).	2+1+2
	OR: (i) Mass of reactants equals mass of products (1). (ii) Weigh flasks containing the two solutions, mix them to get a white precipitate of barium sulfate, reweigh; the masses are equal (3). (iii) Escaping gas would carry away mass (1).	1+3+1
38	(i) Longer stopping time gives smaller acceleration (1). (ii) a = 300 m s^{-2}, F = 45 N (2). (iii) a = 3000, F = 450 N (2).	1+2+2
	OR: (i) A body continues in rest or uniform motion unless a net force acts (1). (ii) With less friction a body goes farther; with none it would move forever (2). (iii) Passenger lurching when the bus stops; dust shaken off a carpet (2).	1+2+2
39	(i) Into X and out of Y (1). (ii) The solution around Y (1). (iii) The rigid cell wall resists the pressure (1). (iv) Plasmolysis (2).	1+1+1+2
	OR: (i) Fluid mosaic of lipids and proteins (2). (ii) Selectively permeable (1). (iii) Osmosis is the diffusion of water across a membrane; diffusion needs no membrane (2).	2+1+2
40	(i) Diagram (2). (ii) Sepals protect, petals attract, stamens make pollen, pistil contains ovules (2). (iii) Unisexual: only stamens or pistils; bisexual: both (1).	2+2+1
	OR: (i) Cross-pollination (1). (ii) Wind (1). (iii) Gives variation and hybrid vigour (1). (iv) Anther and stigma (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER